

LABWORK 2

*This lab aims to teach **how to call a function inside another function** and how to **transfer data between functions**.*

Task 1: Cylinder Volume and Surface Area Calculator (30 pts)

Learn to use functions by calculating the volume and surface area of a cylinder.

Function 1:

- **Name of the function:** `calculate_circle_area`
- **Return type:** `double`
- **Input parameters:** 1 (`double radius`)
- **Description:** Calculates and returns the area of a circle given its radius. (Use 3.14159 for π .)

Function 2:

- **Name of the function:** `calculate_circle_circumference`
- **Return type:** `double`
- **Input parameters:** 1 (`double radius`)
- **Description:** Calculates and returns the circumference of a circle given its radius. (Use 3.14159 for π .)

Function 3:

- **Name of the function:** `calculate_cylinder_volume`
- **Return type:** `double`
- **Input parameters:** 2 (`double radius`, `double height`)
- **Description:** Calculates and returns the volume of a cylinder. This function must call `calculate_circle_area` to get the base area. (Volume = Base Area * Height)

Function 4:

- **Name of the function:** `calculate_cylinder_surface_area`
- **Return type:** `double`
- **Input parameters:** 2 (`double radius`, `double height`)
- **Description:** Calculates and returns the total surface area of a cylinder. This function must call `calculate_circle_area` to get the base area and `calculate_circle_circumference` to get the base circumference for the lateral surface area. (Total Surface Area = 2 * Base Area + Lateral Surface Area; Lateral Surface Area = Base Circumference * Height)

Main Function:

- **Name of the function:** `main`
- **Return type:** `int`
- **Input parameters:** none

- **Description:**

1. Request two inputs from the user: one for the cylinder's radius and the other for its height (both as doubles).
 2. Call the `calculate_cylinder_volume` function to determine the cylinder's volume.
 3. Call the `calculate_cylinder_surface_area` function to determine the cylinder's total surface area.
 4. Print the calculated volume and surface area to the screen.
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Task 2: Electricity Bill Calculator (70 pts)

Learn to use functions within other functions by calculating an electricity bill based on age, consumption, and house type.

Scenario

A user's electricity bill is calculated according to:

- **House Type:** Apartment (A) or Villa (V)
- **Age:** (for discount)
- **Monthly Consumption (kWh)**

The final bill depends on:

1. The base unit price (per kWh)
 2. The consumption multiplier
 3. Discount or extra charge depending on the user's age
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Function 1:

- **Name:** `getBaseRate`
- **Return Type:** `double`
- **Input Parameters:** `char houseType`
- **Description:**

Returns the base price per kWh:

- Apartment (A/a): 0.85 TL/kWh
 - Villa (V/v): 1.10 TL/kWh
-

Function 2:

- **Name:** `calculateConsumptionCost`
- **Return Type:** `double`
- **Input Parameters:** `double baseRate, double kWh`
- **Description:**

Calculates the cost before discounts or extra charges:

`cost = baseRate * kWh`

Function 3:

- **Name:** `applyAgeAdjustment`
 - **Return Type:** `double`
 - **Input Parameters:** `int age, double cost`
 - **Description:**
 - If $\text{age} < 18 \rightarrow 10\%$ discount
 - If $18 \leq \text{age} < 65 \rightarrow$ no change
 - If $\text{age} \geq 65 \rightarrow 15\%$ discountReturns the adjusted cost.
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Function 4:

- **Name:** `calculateTotalBill`
 - **Return Type:** `double`
 - **Input Parameters:** `int age, char houseType, double kWh`
 - **Description:**

This function **calls all other functions**:

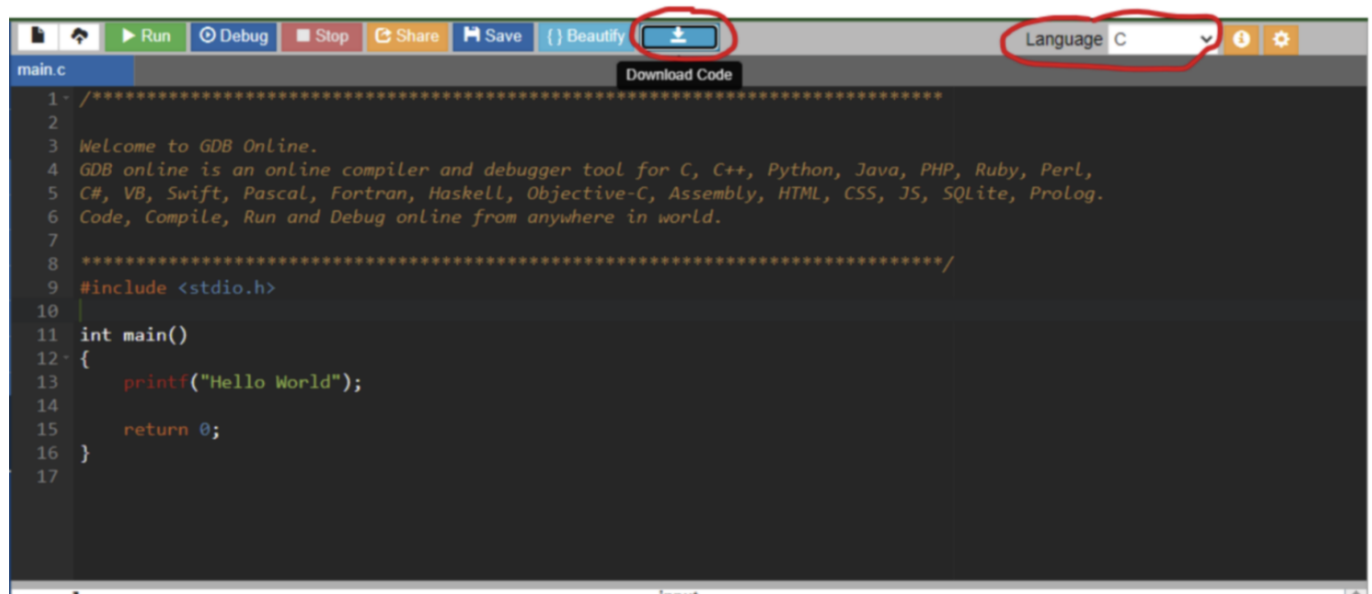
 1. Calls `getBaseRate` to find the unit price
 2. Calls `calculateConsumptionCost` to find base cost
 3. Calls `applyAgeAdjustment` to find the final billReturns the total amount.
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Main Function:

- **Name:** `main`
 - **Return Type:** `int`
 - **Input Parameters:** none
 - **Description:**
 - Asks the user for their age, house type, and monthly consumption (kWh)
 - Calls `calculateTotalBill`
 - Prints the total bill amount
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Restrictions

- If you are not sure something is free to use or not, please ask your assistant.
- You have to complete the labwork using the functions provided. If you complete the labwork without using functions, your work will not be graded.
- Mobile phone and internet usage are not allowed.
- You can only access yulearn and the online C compiler <https://www.onlinegdb.com/>
- Do not forget to select the language if you use onlinegdb. When you finish your work, you can download your code by using the download code button on top of the window.



```
1  /******  
2  
3  Welcome to GDB OnLine.  
4  GDB online is an online compiler and debugger tool for C, C++, Python, Java, PHP, Ruby, Perl,  
5  C#, VB, Swift, Pascal, Fortran, Haskell, Objective-C, Assembly, HTML, CSS, JS, SQLite, Prolog.  
6  Code, Compile, Run and Debug online from anywhere in world.  
7  
8  *****/  
9  #include <stdio.h>  
10  
11 int main()  
12 {  
13     printf("Hello World");  
14  
15     return 0;  
16 }  
17
```

Submission

- Submit your C file with the format "name_surname.c" (use your name and surname).
- Do not submit a Word document, text file, or executable (a.out).